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COAL- A2

R = 0x3A

Q1 (a)-

ax → FFFF
bx → 003A
FFC5

⇒ CF = 0

Hence, jump is not taken.

(b)

1 1 1
ax → 7FFF
R → 003A
8039

⇒ MSB = 1 ⇒ -ve no.
↳ SF → 0 → 1
& OF → 1

Hence, jump is taken

(c)

cx → 00FF
R → 003A
00C5 ≠ 0000

Hence, jump is not taken.

(d)

al → 3A → MSB → 0
+ → 45 → SF → 0
9F → OF → 0

Hence, jump is not taken

(e) → (d) If it was 4A or 55 then result would've been SF ⇒ MSB → 1 & SF → 1 which would've switched OF → 1. Hence it would've taken the jump.

Q2.

In little-endian, LSB → lowest address like nums+0 & vice versa.

Address	nums+0	nums+1	num+2	numst3	big+0	big+1	big+2	big+3
Initial	34	12	78	56	DD	CC	BB	AA
After mov word [big+2], 0x9999	34	3A (al)	00 (ah)	DD (big+word → Lower byte)	CC (→ upper byte)	CC	99 (mov)	99 (9999h.)

(b) → In ax → al = 3A & ah = 00 after mov ax, [big] ⇒ means mov word of big → [Little-endian rule → LSB → first → DDCC
hence ax → al = DD & ah = CC.

→ If a third new variable is introduced b/w nums & big, big would shift down in memory & [num+3] ax → will mov CC in that variable while big will produce same output for rest.

CamScanner

big → DD CC 99 99.

Q3 (a) al → B0 + 0A = BA CF = 0, 1. rcl al, 1 ⇒ 1011 1010 ← 0 = 0111 0100 ↳ CF = 1

2. ror al, 2 ⇒ 0111 0100 ③ shl al, 1

→ 0011 1010, ① ⇒ CF = 0 ↳ 0001 1101 ← 0

0001 1101 ② ⇒ CF = 0 al = 0011 1010, CF = 0

4. rol al, 3 ⇒ 0011 1010 ⇒ 0111 0100 CF = 0 ①

1110 1000, CF = 0 ②, 1101 0001, CF = 1 ③

5. sh r al, 1 ⇒ 0 → 1101 0001 = 0110 1000, CF = 1

Hence AL = 68h, CF = 1.

(b) Reconstruction: → 0110 1000 CF = 1

→ In last instruction, we shifted al by 1 to the right.

→ If we do shl al 1 → 1101 0000 → we now get D.

⇒ For 1, if we rcl al, 1 ⇒ 1101 0001.

Hence, 1D×0 ⇒ rcl al, 1.

Q4 Execution

